Welcome to the **Fall 2026**

Blockchain Innovation Challenge powered by Stellar

Kick-off and Team Formation

November 2





UNIVERSITY OF WATERLOO | VELOCITY | Stellar

Your host for the evening & #1 fan

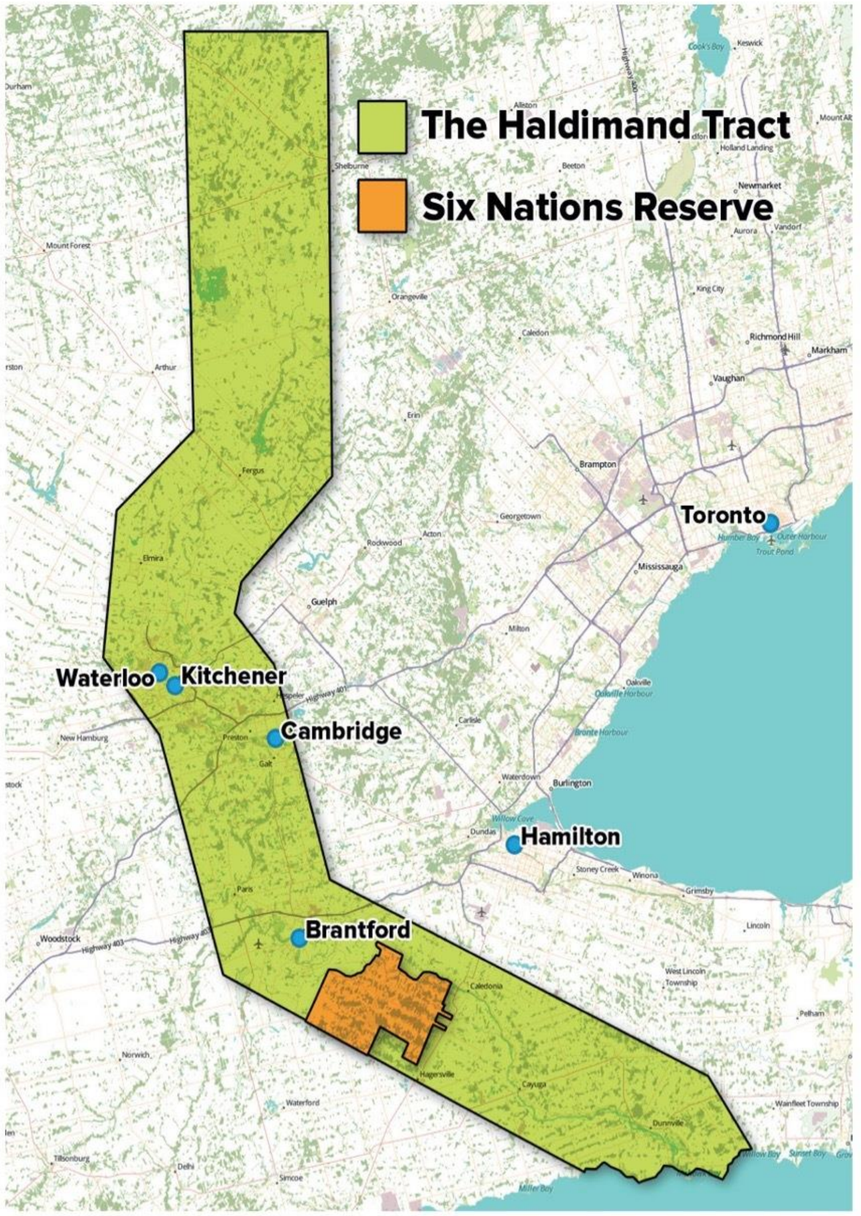


Krysta Traianovski

Associate Director, Founder Development, Velocity

Territorial Acknowledgement

The University of Waterloo acknowledges that much of our work takes place on the traditional territory of the Neutral, Anishinaabeg and Haudenosaunee peoples. Our main campus is situated on the Haldimand Tract, the land granted to the Six Nations that includes six miles on each side of the Grand River. Our active work toward reconciliation takes place across our campuses through research, learning, teaching, and community building, and is co-ordinated within the Office of Indigenous Relations.



UNIVERSITY OF WATERLOO | VELOCITY

Agenda

- Introduction to the Blockchain Innovation Challenge powered by Stellar and overview of problem spaces | 4:15 - 5:10 PM
- Q&A | 5:10-5:20 PM
- Team formation | 5:20-6:00 PM
- Closing remarks | 6:00 PM

Meet the Velocity Events team!



Val
Events Manager



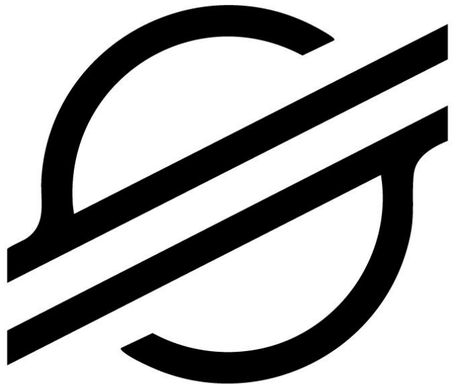
Rachel
Events Coordinator

UNIVERSITY OF WATERLOO | VELOCITY



De-Ann Abraham

Global Engagement Lead , Stellar Development Foundation



Stellar

The Blockchain Innovation Challenge powered by Stellar

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Stellar
Development Foundation

UNIVERSITY OF WATERLOO · VELOCITY

The Blockchain Innovation Challenge

powered by Stellar



Stellar

What is Stellar?



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Stellar

The blockchain for financial services

A public, configurable network with 12 years of proven performance and institutional-grade controls.

Treasury operations	B2B payments & settlements
Payroll	Lending & collateral
Remittances	Aid and benefits disbursement
Savings & investing	Trading & market making

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Stellar

Why institutions choose Stellar



Proven reliability

99.998% uptime since 2014; 100% uptime on a rolling five-year basis. Publicly identifiable validators reach consensus by agreement — no mining or staking.



Native asset controls

Configure issuer controls like authorization, freeze, and clawback directly at the protocol layer — no smart contracts required.



Global distribution

Connect issued assets to wallets, exchanges, payment rails, and on/off ramps around the world, including MoneyGram in 170+ countries.



Expanding ecosystem

Monthly active developers grew 161% YoY, outpacing other major networks and accelerating the growth of applications and infrastructure on Stellar.



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Stellar

The Stellar Consensus Protocol

Institutional-grade consensus

Built on trust

Validators are vetted institutions, not anonymous computers. Transactions clear only when trusted parties agree, preventing bad actors and conflicting records.

Resilient by design

Distributed validator groups provide redundancy if individual participants go offline. Transactions settle in seconds, 24/7.

Predictable fees

Validation is based on agreement, not competition, so fees stay at fractions of a cent. No bidding wars, no surprises. Costs you can model, plan around, and rely on.

Sanctions-safe by design

No validator rewards. No mining incentives. Institutions choose which validators they trust — giving them direct control over their consensus relationships.



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Stellar

Built for regulated finance



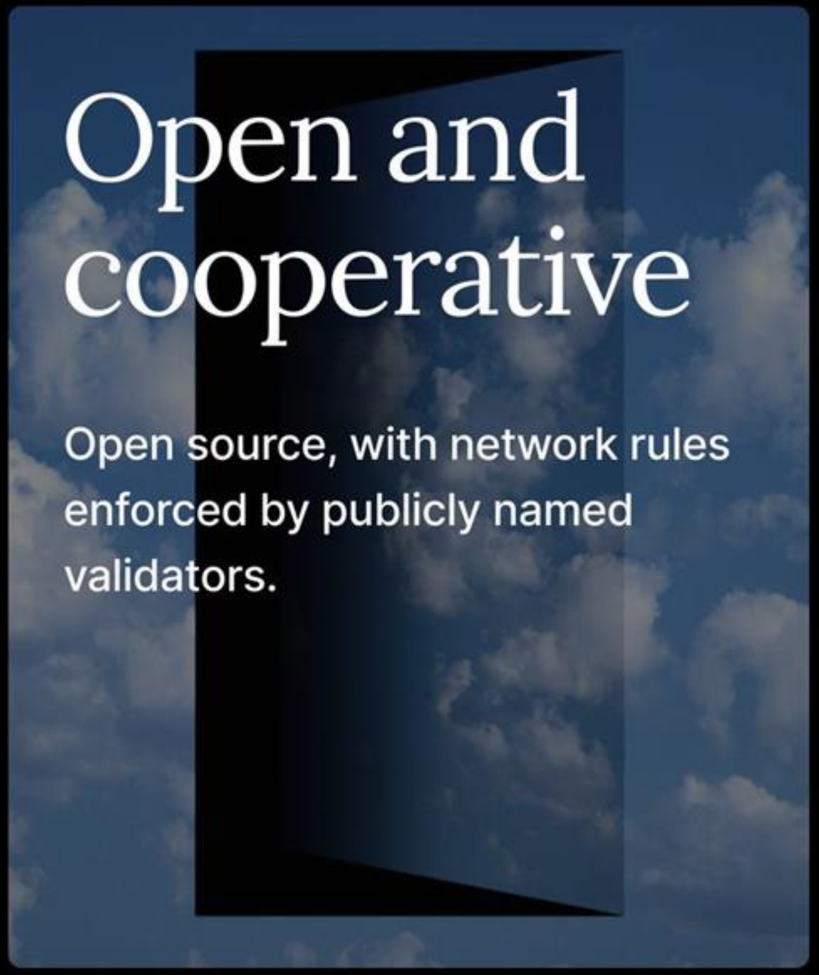
Configurable by design

Compliance and interoperability features tailored to any use case.



Proven since 2014

Operating in production for 12+ years, battle-tested for institutional finance.



Open and cooperative

Open source, with network rules enforced by publicly named validators.

*https://stellar.org/blog/developers/may-15th-network-halt **https://www.helius.dev/blog/solana-outages-complete-history *Uptime per SDF network data, as of Aug 2026*

Stellar

Network performance

3,351

Theoretical maximum TPS

How many transactions per second Stellar is theoretically capable of.

Chainspect, Aug 2026

356

Mainnet max TPS

Highest sustained transaction per second (1 block) on mainnet.

Chainspect, Aug 2026

~5s

Block close time

Average block generation latency under decentralized consensus.

2026 YTD

~$0.0002

Average txn fee

Predictable fee structure, even at high volumes.

2026 YTD

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Stellar

Real users, real results

usbank · PayPal · Franklin Templeton · Circle · DTCC (COMING 2027) · WisdomTree · MoneyGram · spiko

24B+
Lifetime network operations

$9.9B+
Quarterly stablecoin payment volume (Q2 2026)

$4B+
Real-world assets on Stellar (excl. stablecoins)

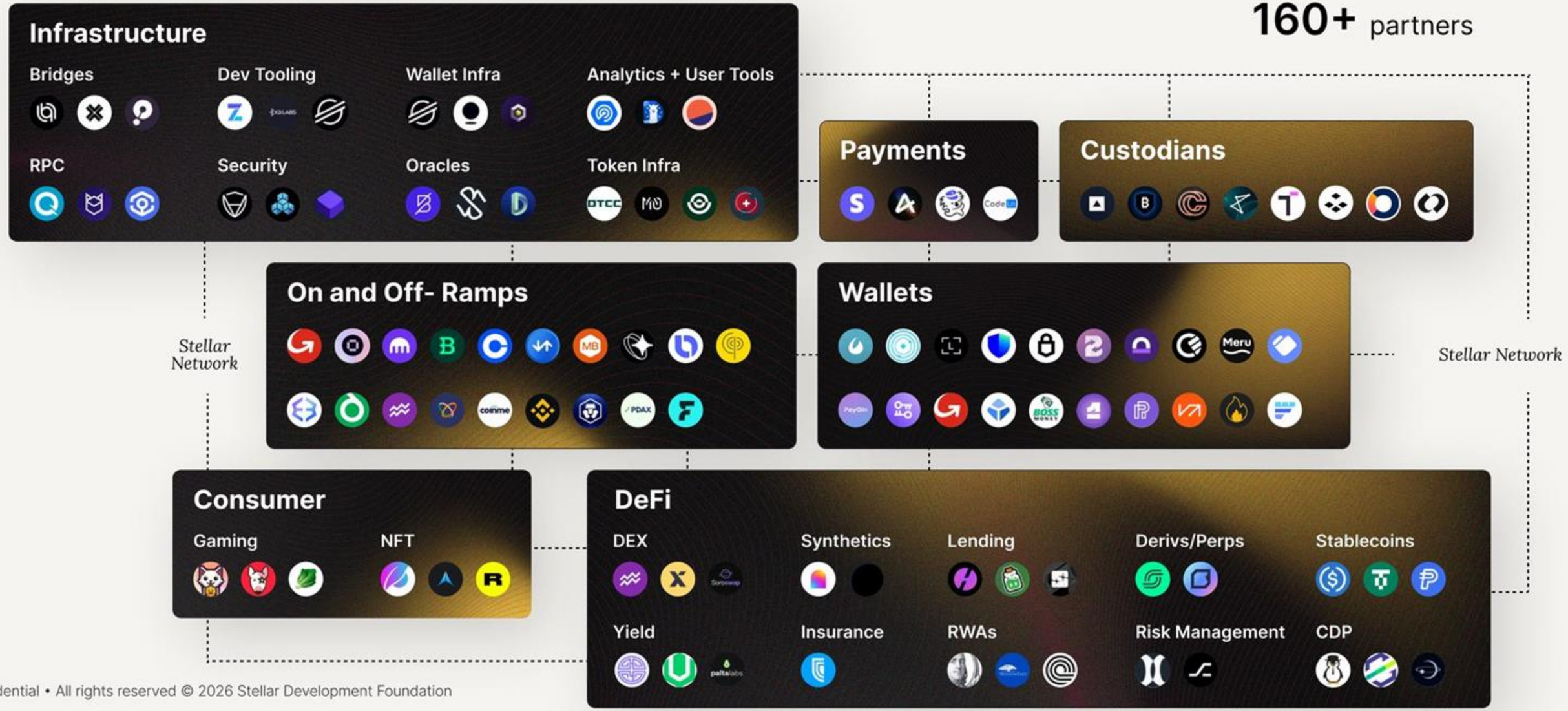
170+
Countries with Stellar activity

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Stellar

An expanding financial ecosystem

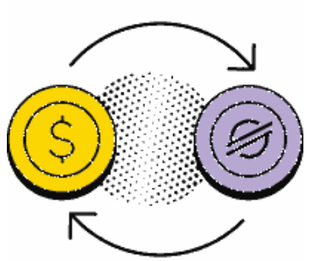
160+ partners



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THE OPPORTUNITY

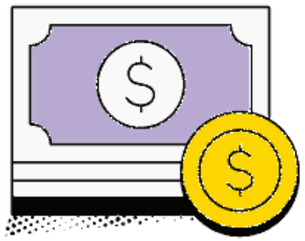
Everyday financial services remain out of reach for billions of people



Payments



Savings



Credit



Insurance



Asset ownership

The challenge targets everyday financial services (EDFS). All three problem spaces work on making these services reach the people who still go without them.

THE GAP

Account ownership reached 79%, yet 1.3 billion adults remain unbanked

79%

of adults now hold a financial account, up from 74% in 2021

1.3B

adults remain unbanked, over half of them in just eight countries

55%

of the unbanked are women; 52% come from the poorest 40% of households

~900M

unbanked adults already own a mobile phone; about 530M a smartphone

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Account access alone is insufficient. Many account holders still cannot affordably send money abroad, protect earnings from inflation, obtain credit, or invest.

HOW TO APPROACH THE CHALLENGE

Strong solutions make serving underserved users structurally cheaper



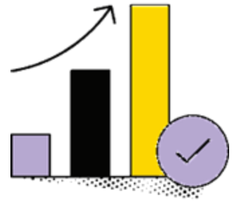
Start with a customer and a problem

Each problem space names a customer and a problem rather than an architecture.



Build on Stellar where money moves

Where a solution touches money movement, digital assets or settlement, build on Stellar: payments, assets, anchors and ramps, and Soroban.



Make the unit economics work

Make it cheaper, so it works at market prices. Stellar's speed and near-zero cost help.

THE PROBLEM SPACES

Three problem spaces.



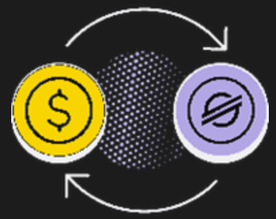
01

Access to Fair Financial Assets and Affordable Financing



02

Verification Infrastructure for the Next Billion Customers



03

Payments: Cross-Border Remittances, Transnational Households, and the Last Mile

01



PROBLEM SPACE

Access to Fair Financial Assets and Affordable Financing

THE PROBLEM

For billions of people, the hard problem is keeping money and putting it to work

Saving

- Inflation erodes savings faster than accounts pay
- Fees and paperwork push savers into cash or goods
- Tokenized assets rarely reach everyday products

Why it matters

Savings lose value to inflation within weeks.

Borrowing

- Underwriting costs more than a small loan earns
- Rigid schedules turn seasonal cash gaps into defaults
- Predatory terms and hidden fees start debt spirals

Why it matters

Financing is harder still for small business owners and informal workers.

Women

- Save heavily in informal channels, yet are least served formally
- Often lack formal title to the assets they use daily

Why it matters

Formal accounts exclude low-income savers with minimums and paperwork.

THE PROBLEM

The financing gap runs to trillions, and much borrowing still happens between family and friends

$5.7T

MSME finance gap in emerging and developing economies, 19% of their GDP

$1.9T

of that gap belongs to women-owned businesses, about a third of the total

31%

of adults in developing economies borrowed from family or friends; 24% borrowed formally

40%

of adults in developing economies saved formally in 2024, up 16 points since 2021

Demand is proven and formal saving is rising fast. The gap is a cost problem: lenders can't cheaply see, value or recover what borrowers hold.

Sources: IFC, MSME Finance Gap (2024, based on 2019 data); World Bank, Global Findex Database 2025 (2024 survey).

OPPORTUNITIES

Informal businesses are barely counted: the $5.7T estimate rests on 2019 data.

Opportunity → a verifiable record of sales, stock and repayments that lenders trust.

Informal assets have no price history, from a stored harvest to a used motorbike.

Opportunity → valuation, claim and recovery models for the collateral people hold.

Savings groups move real money but leave little data behind.

Opportunity → digitize the chama or tanda without breaking the trust that makes it work.

Real yields on small balances, after fees and inflation, are rarely published.

Opportunity → products that show savers exactly what they keep.

Sources: IFC (2024); SME Finance Forum, MSME Finance Gap; World Bank, Global Findex Database 2025.

TARGET AREAS

Five target areas

1	**Everyday stores of value**	How might we let a saver in a high-inflation economy hold dollars or tokenized T-bills from $10, and cash out the same day?
2	**Micro-savings and community savings**	How might we bring micro-savings, village savings and loans onchain without losing the trust that makes it work?
3	**Credit for productive use and resilience**	How might we finance a month of inventory for a market trader, or an emergency, at a fair all-in price?
4	**Collateral people actually hold**	How might we turn a stored harvest, a motorbike or an invoice into collateral a lender can verify and recover?
5	**Loans that fit real income**	How might we let repayments flex with seasonal and irregular income, and stay enforceable across borders?

REAL-WORLD CONTEXT

Already happening

On Stellar

Franklin Templeton’s BENJI

Since 2021, the first U.S.-registered money market fund to use a public blockchain as its official record.

On Stellar

Etherfuse stablebonds

Local-currency tokens backed by short-term government bonds such as Mexico’s CETES, bought from dollars for about 0.04%.

Beyond Stellar

Mobile-money credit in Africa

32% of adults in Kenya borrowed through mobile money in 2024, and 22% in both Ghana and Uganda.

Sources: Stellar Development Foundation case studies (Franklin Templeton, Etherfuse); World Bank, Global Findex Database 2025.

PROBLEM SPACE

02

Verification Infrastructure for the Next Billion Customers



Identity, eligibility, and trust, established once and reused

THE PROBLEM

Roughly one billion people cannot prove who they are to a financial institution

Cost and duplication

- $5 to $30 per user makes good customers unprofitable
- Every provider re-verifies the same person from scratch

Exclusion and surveillance

- Rules built elsewhere reject real applicants
- Verification can slide into surveillance
- Women more often lack the documents assumed

Compliance and adoption

- KYC/AML is non-negotiable: aim for proportionate disclosure
- A credential is worth little until many institutions accept it

Why it matters

Hundreds of millions more

have documents but no financial history.

Institutions are blocked by cost

rather than indifference, and every provider repeats the same work.

Software agents now transact

for people, and their identity and authorization must be proven too.

THE PROBLEM

800 million people have no official ID, and billions more can’t use one online

800M+

people have no official proof of identity, down from 850M in 2021; over half are children

2.8B

people lack access to an online digital ID, down from 3.3B in 2021

$0.50

estimated KYC cost per customer in India after e-KYC, down from about $15 on paper

---

Reuse is the lever: verification done once and accepted many times can turn a $5–30 cost into cents.

Sources: World Bank, ID4D–Global Findex 2025; Center for Global Development (2019), citing an Indian Ministry of Finance estimate.

OPPORTUNITIES

Nobody measures false rejections: banks rarely learn which real applicants they turned away.

Opportunity → tools that measure, explain and appeal rejections.

ID counts aren’t KYC counts: having an ID doesn’t mean a bank will accept it.

Opportunity → bridge national/ Government Issued ID to reusable, KYC-ready credentials.

AI Agents that pay on someone’s behalf have no settled identity standard yet.

Opportunity → spending policies and proofs of authority for agents.

Sources: World Bank, ID4D–Global Findex 2025; SDF problem-space brief.

TARGET AREAS

Five target areas

1	**Verification without documents**	How might we verify someone with no ID, address or credit file, to a standard a regulator accepts?
2	**Portable, reusable credentials**	How might we let one verification travel across providers and borders?
3	**Alternative signals**	How might we turn payment, mobile, utility and commerce history into evidence of identity, income and reliability?
4	**Proving a single fact**	How might we prove “over 18”, “resident” or “not sanctioned” without revealing the data behind it?
5	**Recovery and delegation**	How might we restore access after someone loses both phone and documents, or let a family member act for them?

REAL-WORLD CONTEXT

Already happening: the building blocks exist

Beyond Stellar

India’s Aadhaar e-KYC

Reusable digital verification cut the estimated cost of onboarding a customer from about $15 to $0.50.

Beyond Stellar

MOSIP

An open-source national ID platform adopted by countries including the Philippines, Ethiopia and Morocco.

Beyond Stellar

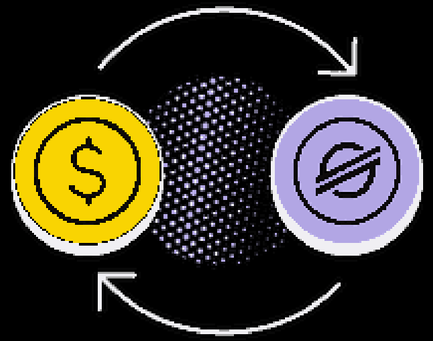
EU Digital Identity Wallet

Each member state must offer one by late 2026; banks and other regulated services must accept it a year later.

PROBLEM SPACE

03

Payments: Cross-Border Remittances, Transnational Households, and the Last Mile



Shortening the distance between a decision and money in hand

THE PROBLEM

High remittance costs, last-mile distribution failures, and institutional delays leave transnational households underserved at both ends of the transfer

Cost

- Cutting total cost from about 6% toward 3%
- Much of the margin hides in the exchange rate

The last mile

- Agents run dry, sit too far away, or go offline
- Delivery must work on any phone or via a cash agent

Institutions

- Weeks pass between approval and money in hand
- Verification must not become surveillance
- Long procurement and security reviews

Why it matters

Transnational households are feel the burden more acutely.

Sending money home costs ~6% on average, with extra cost hidden in FX rates.

Institutions pay the same households, yet weeks pass before cash arrives.

THE PROBLEM

Remittances outweigh foreign investment and aid combined, yet cost double the global target

$685B

in remittances to low- and middle-income countries in 2024, more than FDI and aid combined

6.36%

global average cost of sending money home, against an SDG 10.c target below 3%

$2T+

flowed through mobile money in 2025, across 2.3 billion registered accounts

$6.6B

in humanitarian cash and voucher aid in 2024, down from $7.8B as funding fell

Sources: World Bank, People Move (2024) and Remittance Prices Worldwide (2025); GSMA, State of the Industry 2026; CALP Network in ALNAP GHA 2025.

OPPORTUNITIES

Price data stops at the sender: it tracks the cost of sending, not of cashing out.

Opportunity → measure and cut what the recipient pays, in cost and in time.

Informal flows go uncounted, so the true remittance total is likely higher.

Opportunity → serve the corridors official data misses.

Agent float and cash-out failure rates are mostly proprietary.

Opportunity → liquidity forecasting and rebalancing for agents.

Little data tracks whether remittances become savings, insurance or school fees.

Opportunity → products built on cross-border income history.

Sources: World Bank, Remittance Prices Worldwide (367 corridors, 48 sending and 105 receiving countries); World Bank, People Move (2024).

TARGET AREAS

Five target areas

1	**Cash-in, cash-out and agent liquidity**	How might we make sure a rural agent has cash on hand when hundreds of remittances land on the same payday?
2	**Products for the receiving household**	How might we turn a monthly remittance into savings, insurance, credit or a school fee paid directly?
3	**Transparent all-in pricing**	How might we show the full cost, exchange rate included, before anyone hits send, across obligations in two currencies?
4	**Institutional and humanitarian payouts**	How might we get an approved aid payment into a family's hands in hours, not weeks, across bank, mobile-money, agent and wallet rails?
5	**Reconciliation, fraud and reporting**	How might we give donors audit-grade reporting and catch duplication or leakage, without exposing the people they serve?

REAL-WORLD CONTEXT

Already happening: the building blocks exist

On Stellar

UNHCR in Ukraine

Stellar Aid Assist has delivered nearly $5M in direct cash assistance to 2,500+ households, cashed out at MoneyGram.

On Stellar

UNHCR in Argentina

Grants to enterprises serving migrant communities saved about 30% of the operating budget versus currency devaluation.

On Stellar

Hospital payroll in northwest Syria

A GIZ and UOSSM pilot paid health workers; every cash-out was processed on the first attempt.

On Stellar

MoneyGram in Colombia

A stablecoin app lets families receive and hold digital dollars, then cash out at MoneyGram locations.

03 · THE PROBLEM

Hundreds of millions of people earn in one country and support a household in another

Definition

Solutions that shorten the distance between a decision and money in hand, and that treat arrival as a beginning rather than an end.

Why it matters

- **Transnational households are underserved** at both ends of the transfer.
- **Sending money home costs about 6%** on average, with more cost hidden in the exchange rate.
- **Institutions pay the same households,** and weeks routinely pass between approval and money in hand.

How you'll build on Stellar

- Open and free: start on testnet today
- Build with the AI suite: **Raven**, **Stellar Skills**, **stellar-build**
- Ship something real that runs on Stellar
- Use what the ecosystem already offers
- Demo it working, and show what's next

START TONIGHT

Join the builders on Discord

discord.gg/stellardev

Stellar

Supporting builders at every stage

The Stellar Community Fund (SCF) offers community-driven grants of up to $150K in XLM for projects building on Stellar, alongside additional funding:

- Marketing grants
- Infrastructure grants
- R&D grants
- Matching Fund
- Stellar Enterprise Fund
- Academic research grants



1

Problem Space

Access to Fair Financial Assets and Affordable Financing: Savings, Credit, and Inflation Resilience

2

Problem Space:
Verification Infrastructure for the Next Billion Customers



Problem Space:

Payments: Cross-Border Remittances, Transnational Households, and the Last Mile

Stellar Blockchain Innovation Challenge: November 2-18

1

Problem Space 1:
Access to Fair Financial Assets and Affordable Financing: Savings, Credit, and Inflation Resilience.

2

Problem Space 2:
Verification Infrastructure for the Next Billion Customers

3

Problem Space 3:
Payments: Cross-Border Remittances, Transnational Households, and the Last Mile Short Descriptions.

Deliverables

Project Showcase (Monday, November 16th)

1. **A trifold board visual representation** of team's solution.
2. **A three-minute verbal presentation** delivered by one member of the team detailing the solution they have built. The presentation is required to include a one-minute demo video that demonstrates the functioning prototype. Slide-only videos do not meet this requirement. The demo video should be hosted at a publicly accessible link (e.g., YouTube or Loom).
3. Teams must submit a link to a **publicly accessible GitHub repository** containing the complete project source code and a README that describes the project and provides setup and run instructions. Upload to the Blockchain Innovation Challenge Devpost.
4. Teams must deploy a **functional prototype on Stellar Testnet** and provide the deployed contract ID(s) and/or a live application URL so judges can verify and interact with it.

Deliverables

Final Presentation (top eight teams present on Wednesday, November 18th)

1. **A three-minute verbal presentation and Powerpoint slideshow** delivered by one member of the team detailing the solution they have built.
2. Teams must submit a link to a **publicly accessible GitHub repository** containing the complete project source code and a README that describes the project and provides setup and run instructions. Upload to the Blockchain Innovation Challenge Devpost.
3. Teams must deploy a **functional prototype on Stellar Testnet** and provide the deployed contract ID(s) and/or a live application URL so judges can verify and interact with it. Upload to the Blockchain Innovation Challenge Devpost.

Challenge overview

Week 1

Happening now:

- Kick-Off & Team Formation: November 2 | 4:00-6:00 PM | SLC, Black and Gold Room

Tomorrow:

- Intro to Stellar Workshop: Nov 3 | 10:00-11:00 AM | (Virtual – Teams)
- Problem Space Workshops: Nov 3 | 3:00-4:00 PM | (Virtual - Teams)

Wednesday:

- Mentor Roundtables: Nov 4 | 11:00 AM - 1:00 PM | (In Person working space in PSE 7303/7363, advisors will be virtual on Teams)

Thursday:

- Working Session #1: Nov 5 | 4:00-6:00 PM | (In Person – MC Common Room, MC 2037)

Challenge overview

<u>Week 2</u>

Monday

- 1:1 Advisor Office Hours: Nov 09 | 9:00 AM–5:00 PM | (Virtual - Teams)

(sign-up for a time slot by Thursday November 5th at 5:00 PM)

Tuesday

- Technical Workshop: Nov 10 | 5:00-6:00 PM | (Virtual – Teams)

Wednesday

- Working Session #2: Nov 11| 5:00-6:00 PM | (1st floor IDEAS Clinic, PSE, Formerly E7)
- Pitch Prep Workshop: Nov 11| 6:00-7:00 PM | (1st floor IDEAS Clinic, PSE, Formerly E7)

Friday

- One Page Solution Overview DUE: Nov 13 | 8:30 AM Online

Challenge Overview

<u>Week 3</u>

Monday

- Project Showcase: Nov 16 | 4:00-7:00 PM | (2nd Floor Innovation Arena)
  - Needed: trifold board, laptop and three-minute pitch
  - Top eight teams selected from all presentations to move onto the finals

Tuesday

- Finalist Pitch Prep: Nov 17 | 11:00 AM - 1:00 PM | (2nd Floor Innovation Arena)

Wednesday

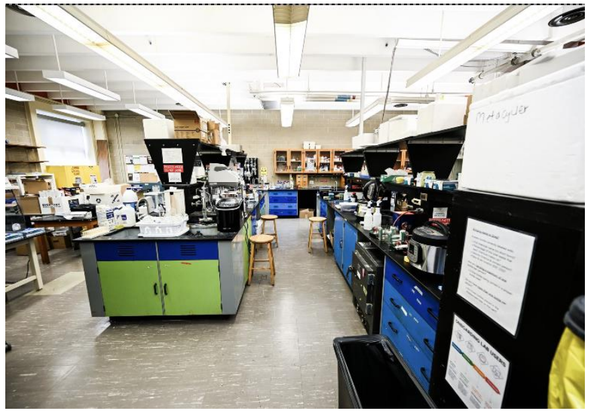
- Final Presentations and Awards: Nov 18| 5:00-7:30 PM | (2nd Floor Innovation Arena)
  - Needed: slideshow and one team member to deliver three-minute pitch
  - Top three teams awarded cash prizes

Many doors into innovation.

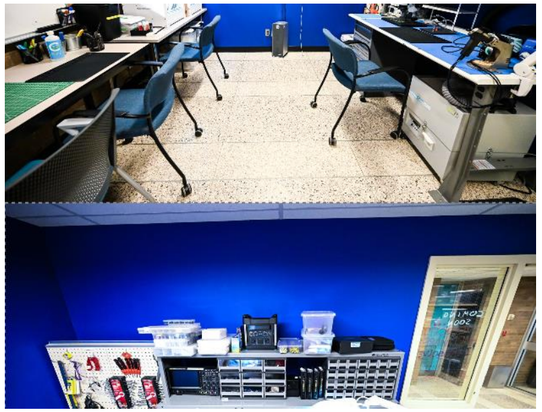
Refer to the Innovation Challenge Notion to access the spaces and find out more details about the specific resources available to you.



Velocity Common Room



Velocity Science



Velocity Hardware



Velocity Digital

Upcoming Events & Stellar Blockchain Innovation Challenge Notion

Tomorrow’s events:

- Intro to Stellar Workshop: Nov 03 | 10:00-11:00 AM | (Virtual – Teams)
- Problem Space Workshops: Nov 03 | 3:00-4:00 PM | (Virtual - Teams)

Scan the QR code to access the Challenge Notion page!

Please take the time to read through it carefully, as it contains essential information, resources, and guidelines to help your team succeed throughout the challenge!


Q
A

Team Formation

You have time to explore 2 problem spaces and connect with peers to form your team.
10 minutes per problem space.

Teams should have 3-4 members.

Each team needs at least one technical participant.

Areas of expertise:

Project Management, Business, UX/UI Design

Blockchain Experience

Accounting & Finance, Mathematic Modelling

Software Development

REGISTER YOUR TEAM!

Before leaving tonight, do you have:

1. **A registered team with at least one technical member**
2. **A trifold board for your team**